# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–5: **Patterns in Mathematics, Lines and Angles, Number Play, Data Handling and Presentation, Prime Time** | Class 6 Maths | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 15 × 1 = 15

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** 17 | The common difference is 3, so add 3 to the last term. |
| 2 | **(a)** acute | Compare $30^\circ$ with $90^\circ$ and $180^\circ$. |
| 3 | **(a)** 8531 | Write the digits in decreasing order. |
| 4 | **(a)** 30 | Multiply the symbols by the value of one symbol. |
| 5 | **(c)** 13 | 13 has only two factors, 1 and 13. |
| 6 | **(b)** 15 | The $n$th odd number is $2n - 1$. |
| 7 | **(c)** 60 | The hour hand turns $30^\circ$ in 1 hour. |
| 8 | **(d)** 1358 | Write the digits in increasing order (a non-zero digit first). |
| 9 | **(c)** 35 | Multiply the height by the scale. |
| 10 | **(a)** 27 | 27 has more than two factors. |
| 11 | **(b)** 14 | The $n$th even number is $2n$. |
| 12 | **(a)** 90 | A full turn is $360^\circ$. |
| 13 | **(b)** 454 | A palindrome reads the same forwards and backwards. |
| 14 | **(d)** 1 full bundles and 2 extra stroke(s) | Every fifth stroke crosses the four before it, making a bundle of five. |
| 15 | **(d)** 6 | Factors: 1, 2, 3, 4, 6, 12. |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 16 | Common difference $= 5$ (1); Next three: $24, 29, 34$ (1). | 1+1 |
| 17 | It is acute (1); Compare with the right angle $90^\circ$ (and the straight angle $180^\circ$) (1). | 1+1 |
| 18 | Largest $9752$ (1); Smallest $2579$ (1). | 1+1 |
| 19 | Each symbol is 10 students (1); $6 \times 10 = 60$ students (1). | 1+1 |
| 20 | Find pairs whose product is 16 (1); Factors: 1, 2, 4, 8, 16 (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 5 × 3 = 15

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | Common difference $= 2$ (1); Term $n$ is $a + (n-1)d = 3 + 7 \times 2$ (1); 17 (1). | 1+1+1 |
| 22 | $quarter$ of $360^\circ$ (1); $= 90^\circ$ (1); Type: right (1). | 1+1+1 |
| 23 | Largest $= 741$, smallest $= 147$ (1); Sum $= 888$ (1); Difference $= 594$ (1). | 1+1+1 |
| 24 | Add all the numbers (1); Total $= 15 + 35 + 10 + 15 = 75$ (1); Hence the total is 75 (1). | 1+1+1 |
| | **OR:** Divide by primes in turn (1); $36 = 2 \times 2 \times 3 \times 3$ (1); Only prime factors appear (1). | 1+1+1 |
| 25 | $1 + 3 + 5 + 7 + 9 = 25$ (1); $25 = 5 \times 5$ is a square number (2). | 1+2 |
| | **OR:** Each hour mark is $30^\circ$ apart (1); Angle $= 1 \times 30 = 30^\circ$ (smaller angle $30^\circ$) (1); Type: acute (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 26 | First 6 terms: $2, 5, 8, 11, 14, 17$ (1); Term 7 is 20 and term 8 is 23 (2); Yes, 23 is the 8th term (2). | 1+2+2 |
| | **OR:** $30^\circ$ is acute (1); $360 \div 30 = 12$ (1); 12 turns make a full turn (1); Check: product is $360^\circ$ (2). | 1+1+1+2 |
| 27 | Total $= 1245 + 1378 = 2623$ (1); Nearest hundred: 2600 (2); Estimate is close to the actual value (2). | 1+2+2 |
| | **OR:** A: $4 \times 10 = 40$ (1); B: $6 \times 10 = 60$ (1); C: $5 \times 10 = 50$ (1); Total $= 150$ (2). | 1+1+1+2 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–5: **Patterns in Mathematics, Lines and Angles, Number Play, Data Handling and Presentation, Prime Time** | Class 6 Maths | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 15 × 1 = 15

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** 36 | $6 \times 6 = 36$. |
| 2 | **(b)** 3 | Count every pair of points. |
| 3 | **(b)** 1000 | Add 1. |
| 4 | **(c)** Guava | Look for the largest number. |
| 5 | **(b)** 42 | $7 \times 6$ |
| 6 | **(a)** 10 | $\frac{4\times5}{2} = 10$. |
| 7 | **(d)** two end points | Definition |
| 8 | **(c)** 350 | Look at the ones digit: 5 or more rounds up. |
| 9 | **(a)** Tennis | Look for the smallest number. |
| 10 | **(b)** 363 | Divisibility test of 3. |
| 11 | **(d)** 8 | $2 \times 2 \times 2 = 8$. |
| 12 | **(b)** 30 | In 1 minute the minute hand turns $6^\circ$. |
| 13 | **(a)** 3500 | Look at the tens digit. |
| 14 | **(c)** 4 | Count (tally) the number of times. |
| 15 | **(c)** 10 | Add the primes 2, 3, 5, 7, ... |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 16 | Squares are $1 \times 1, 2 \times 2, \ldots$ (1); $1, 4, 9, 16, 25$ (1). | 1+1 |
| 17 | It turns $30^\circ$ per hour (1); $3 \times 30 = 90^\circ$ (1). | 1+1 |
| 18 | Reverse of 636 is 636 (1); Yes, it is a palindrome (1). | 1+1 |
| 19 | Draw strokes; every fifth stroke crosses four (1); 9 = 1 full bundles and 4 extra stroke(s) (1). | 1+1 |
| 20 | Multiply 6 by 1, 2, 3, 4, 5 (1); 6, 12, 18, 24, 30 (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 5 × 3 = 15

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | Reverse $= 21$ (1); Sum $= 12 + 21 = 33$ (1); Yes, palindrome (1). | 1+1+1 |
| 22 | Count each value (1); 1: 2; 2: 5; 3: 4; 4: 4 (1); Total $= 15$ (1). | 1+1+1 |
| 23 | By 2: no (last digit 5) (1); By 3: yes (digit sum 9) (1); By 5: yes (1). | 1+1+1 |
| 24 | $1 + 2 + \cdots + 9$ (1); $= 45$ (2). | 1+2 |
| | **OR:** It turns $6^\circ$ per minute (1); $8 \times 6 = 48^\circ$ (1); Less than $90^\circ$ (1). | 1+1+1 |
| 25 | 3482 rounds to 3500 and 2619 rounds to 2600 (1); Estimate $= 6100$ (1); Exact $= 6101$ (1). | 1+1+1 |
| | **OR:** Most: Tennis (50) (1); Least: Football (25) (1); Difference $= 25$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 26 | $12 = 2 \times 2 \times 3$ (1); $18 = 2 \times 3 \times 3$ (2); Common prime factors: 2, 3 (2). | 1+2+2 |
| | **OR:** $1 + 3 + 5 + 7 = 16$ (1); $16 = 4 \times 4$ (2); Sum of first 5 odd numbers $= 5 \times 5 = 25$ (2). | 1+2+2 |
| 27 | Segments: $4 \times 3 \div 2 = 6$ (1); Rays from A: one through each of the other 3 points (1); 3 rays (1); Rays can be extended without end, segments cannot (2). | 1+1+1+2 |
| | **OR:** Greatest $= 8642$ (1); Smallest $= 2468$ (1); Difference $= 6174$ (1); Even (2). | 1+1+1+2 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–5: **Patterns in Mathematics, Lines and Angles, Number Play, Data Handling and Presentation, Prime Time** | Class 6 Maths | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 15 × 1 = 15

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** 16 | The sum of the first $n$ odd numbers is $n \times n$. |
| 2 | **(a)** right | Compare $90^\circ$ with $90^\circ$ and $180^\circ$. |
| 3 | **(c)** even | Rule |
| 4 | **(a)** data | Definition |
| 5 | **(a)** 2 | Definition |
| 6 | **(c)** 25 | The common difference is 4, so add 4 to the last term. |
| 7 | **(c)** 90 | The hour hand turns $30^\circ$ in 1 hour. |
| 8 | **(c)** 8 | Count places from the right: ones, tens, hundreds, thousands. |
| 9 | **(d)** 70 | Multiply the symbols by the value of one symbol. |
| 10 | **(d)** 29 | 29 has only two factors, 1 and 29. |
| 11 | **(c)** 23 | The $n$th odd number is $2n - 1$. |
| 12 | **(b)** 180 | A full turn is $360^\circ$. |
| 13 | **(a)** 9642 | Write the digits in decreasing order. |
| 14 | **(d)** 60 | Multiply the height by the scale. |
| 15 | **(d)** 49 | 49 has more than two factors. |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 16 | Add 1, then 2, then 3, ... one at a time (1); $1, 3, 6, 10, 15$ (1). | 1+1 |
| 17 | Each pair of points gives a segment (1); 7 points give 21 segments (1). | 1+1 |
| 18 | Nearest hundred: 4400 (1); Nearest thousand: 4000 (1). | 1+1 |
| 19 | Multiply: $4 \times 10$ (1); 40 students (1). | 1+1 |
| 20 | Factors of 31: 1, 31 (1); Prime: only two factors (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 5 × 3 = 15

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | Test each number for factors (1); 11, 13, 17, 19, 23, 29 (2). | 1+2 |
| 22 | Common difference $= 3$ (1); Term $n$ is $a + (n-1)d = 5 + 6 \times 3$ (1); 23 (1). | 1+1+1 |
| 23 | $half$ of $360^\circ$ (1); $= 180^\circ$ (1); Type: straight (1). | 1+1+1 |
| 24 | Largest $= 953$, smallest $= 359$ (1); Sum $= 1312$ (1); Difference $= 594$ (1). | 1+1+1 |
| | **OR:** Add all the numbers (1); Total $= 35 + 15 + 25 + 10 = 85$ (1); Hence the total is 85 (1). | 1+1+1 |
| 25 | Divide by primes in turn (1); $48 = 2 \times 2 \times 2 \times 2 \times 3$ (1); Only prime factors appear (1). | 1+1+1 |
| | **OR:** $1 + 3 + 5 + 7 + 9 + 11 = 36$ (1); $36 = 6 \times 6$ is a square number (2). | 1+2 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 26 | Monday 30 (1); Tuesday 50 (1); Wednesday 40 (1); Total 120 books (2). | 1+1+1+2 |
| | **OR:** Factors: 1, 2, 4, 7, 14, 28 (1); Count $= 6$ (1); Sum of factors except 28 $= 28$ (1); Perfect (1); Even (1). | 1+1+1+1+1 |
| 27 | First 6 terms: $4, 9, 14, 19, 24, 29$ (1); Term 7 is 34 and term 8 is 39 (2); Yes, 39 is the 8th term (2). | 1+2+2 |
| | **OR:** $45^\circ$ is acute (1); $360 \div 45 = 8$ (1); 8 turns make a full turn (1); Check: product is $360^\circ$ (2). | 1+1+1+2 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.